**Checked Example Baseline**

- **Circuit Input:** 4 test vectors V1(0,0), V2(0,1), V3(1,0), V4(1,1)
- **Target Faults:** 4 total faults F1, F2, F3, F4
- **Matrix State:**
  - (0,0) detects None
  - (0,1) detects F1
  - (1,0) detects F2, F4
  - (1,1) detects F1, F3

**Manual Metric Calculations:**

- **Coverage:** 4 out of 4 total target faults are present in the matrix. (4 / 4) * 100 = 100%.
- **Density:** There are 4 total positive detection events (1s) across the grid. The maximum possible grid size is 16 cells (4 vectors * 4 faults). (4 / 16) * 100 = 25%.
- **Multiplicity:** F1 is detected by 2 vectors. F2, F3, and F4 are detected by 1 vector each.
- **Essential Vectors:** Vector (1,0) is essential because it is the only vector to detect F2 and F4. Vector (1,1) is essential because it is the only vector to detect F3. Vector (0,1) is not essential because F1 can also be caught by (1,1).
- **Signature Classes:** F2 and F4 are detected by the exact same set of vectors {(1,0)}. Therefore, they form a signature class. F1 and F3 have unique footprints and form their own individual classes.